WORKSHEET · SCRATCH (GR 6–8) ADVANCED: ALGORITHMS LAB

Lesson 11: From Blocks to Python

Module 4 | 45 minutes | Bridging to text code | Grades 6–8 · ages 12–14

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

a. syntax ___ A. Counting positions from 0.
b. indentation ___ B. Python's "else if".
c. elif ___ C. Spaces at the start of a line that group code.
d. zero-based ___ D. The rules for writing code in a language.

2. Match the Scratch block to its Python

Draw a line from each word to its match.

a. set [x v] to (5) ..... A. def greet(name):
b. repeat (10) ..... B. print("Hi")
c. forever ..... C. while True:
d. say [Hi] ..... D. for i in range(10):
e. define greet (name) ..... E. x = 5

3. Plan your code

Before you build it: FizzBuzz: translate the Python on screen into blocks. For i from 1 to 15: say "FizzBuzz" if i mod 15 = 0, else "Fizz" if i mod 3 = 0, else "Buzz" if i mod 5 = 0, else say i (each for 0.2 seconds). Write or draw the blocks you will use.

4. Trace the Python

What does this Python print? Code: total = 0 ↵ for n in [4, 7, 9]: ↵ if n % 2 == 1: ↵ total = total + n ↵ print(total)

a. A. 16
b. B. 20
c. C. 4

5. Python essentials

In Python, a My Block is written with the keyword _____. The blocks inside a loop are shown by _____. The first item of a Python list is at index _____. (Word bank: def, indentation, 0, 1, loop)

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can translate between Scratch and Python.
☐ I can implement an algorithm from Python code.
☐ I can name key Python syntax rules.